

Process Simulation Services

Where Process Meets Intelligence.



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Solutions

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Steady-state simulation



Software: ASPEN HYSYS/ ASPEN PLUS/ PRO II/ PETROSIM/ UNISIM

Why?

In today's competitive industrial landscape, an accurate Heat and Mass Balance (HMB) is not just a technical document—it is the foundation of cost-effective design, safe operations, and long-term profitability. Our team specializes in Aspen HYSYS steady-state simulation, delivering high-fidelity models that go far beyond basic calculations. We transform your raw process data into actionable engineering insights, empowering you to optimize energy consumption, eliminate operational bottlenecks, and confidently make decisions that reduce capital and operating expenses—before you ever spend a dollar on hardware.

Success story:

Client Challenge:

In an Air Separation Unit (ASU), the client wanted to switch production from Medium-Pressure Oxygen (MPO) to High-Pressure Oxygen (HPO). This required increasing upstream pressure feeding the main LNG-type cryogenic heat exchanger.

The Client's Concern:

The client feared this pressure increase would significantly alter the energy balance in the main exchanger, causing severe nitrogen production loss and creating a decision-making bottleneck.

Our Solution & Analysis:

We performed a rigorous steady-state simulation using Aspen HYSYS. A high-fidelity model of the ASU and its main cryogenic exchanger was built, testing multiple scenarios to evaluate the pressure change's impact on the energy balance. The simulation revealed a crucial insight: the liquid stream's enthalpy did not change significantly with increased pressure. The thermal duty in the main exchanger remained stable, meaning the shift to HPO would not cause meaningful nitrogen loss.

The Outcome & Decision Made:

Armed with validated data, the client made a confident decision to proceed with the operating change. By proving the energy balance was stable, we eliminated production risk and enabled the client to capture HPO benefits without compromising nitrogen output.

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Dynamic simulation



Software: ASPEN HYSYS/ UNISIM/ ASPEN Adsim

Why?

Your plant doesn't operate at a single steady-state point; it experiences start-ups, trips, and continuous fluctuations. Relying only on static calculations leaves you blind to critical operational risks. Our dynamic simulation services model the true transient behavior of your process, allowing us to virtually test control narratives, emergency shutdown sequences, and compressor anti-surge systems. The result? You gain the confidence to make critical operational decisions—improving safety, protecting assets, and avoiding costly production downtime.

Success story:

Client Challenge:

In an Air Separation Unit (ASU), the client needed to switch over the Main Air Compressor to a standby unit without disrupting plant operations.

The Client's Concern:

Performing this switchover on the real plant was risky—any fluctuation could trip the entire ASU or push the compressor into a hazardous zone.

Our Solution & Analysis:

We built a dynamic HYSYS model of the compressor train and simulated the complete switchover, monitoring all process variables and their rates of change at every stage. The simulation confirmed the switchover could be performed smoothly—no surge, no trips, no unsafe conditions.

The Outcome & Decision Made:

With validated data, the client confidently implemented the procedure on the real plant, ensuring safe and continuous operation without risk to equipment or personnel.

Software: ASPEN HYSYS/ UNISIM/ JAVA/ Python

Why?

Your plant never truly runs at steady state, and neither should your training. Static procedures don't prepare operators for the chaos of real-world upsets. Our OTS delivers a safe virtual environment with a fully replicated DCS and SIS, featuring a similar HMI—so every screen, alarm, and control button looks and behaves exactly like the real console. Operators train on start-ups, trips, and emergency scenarios repeatedly—without risking the plant. The result? A workforce that responds with confidence, reduces human error, and prevents costly downtime.

Success story:

In an Air Separation Unit (ASU), we built a complete dynamic simulation with all safety loops and replicated the client's HMI/DCS exactly. Operators trained on the same screens and alarms they use daily, building confidence through repeated practice. By simulating power failures, valve failures, and compressor trips, they mastered emergency response in a risk-free environment. The result? A confident, fully prepared workforce that makes critical decisions under pressure—protecting the plant and personnel without risking real operations.

Digital twin

Software: ASPEN HYSYS/ UNISIM/ JAVA/ Python

Why?

When connected online with the real plant instrumentation, the Process Digital Twin becomes a Real-Time Simulator (RTS)—a live digital mirror of your operating asset that continuously calculates pressures, temperatures, flows, and compositions for every stream and every tray in every column. The RTS acts as a soft sensor and backup for online analyzers, provides advanced bad actor detection by creating holistic alarms like "Compressor degraded efficiency" instead of relying on single physical instruments, and delivers look-ahead predictions by running faster than real time to forecast events such as "High pressure column will flood in 10 minutes" or "Product quality will go off-spec in 30 minutes." Because the RTS is always synchronized to the current plant state, it also serves as a what-if advisor—allowing operators and engineers to test actions virtually before executing them on the real plant—and with autosaved snapshots every few minutes, it enables post-incident analysis by restoring exact past conditions, repeating the scenario in simulation, studying the best operator response, and communicating those lessons to the entire team. The result? A plant that is not just monitored, but deeply understood—empowering your team to act proactively, prevent failures, and optimize performance in real time.

RTO



Software: ASPEN HYSYS/ UNISIM/ Python

Why?

Your plant never truly runs at its optimal point—feed quality changes, ambient conditions shift, equipment degrades, and market prices fluctuate. Static design calculations can't keep up with the reality of daily operations. Our Real-Time Optimization (RTO) services use a high-fidelity, rigorous process model that continuously calculates the optimal operating conditions for your plant—maximizing yield, minimizing energy consumption, and improving profitability—while respecting all process constraints, equipment limits, and safety margins. The RTO delivers constraint-aware recommendations that identify opportunities static analysis misses, enables what-if analysis for engineers to test scenarios virtually before implementing changes on the real plant, and uncovers debottlenecking opportunities by quantifying the financial impact of removing process limitations. The result? A plant that doesn't just run—it runs smart, empowering your team to make proactive, data-driven decisions that increase throughput, reduce energy costs, and maximize profitability while maintaining safety and operational stability.

Fault Diagnostic

Software: ASPEN HYSYS/ UNISIM/ Python

Why?

Your plant's critical faults—compressor surge, heat exchanger fouling, reactor runaway, valve sticking, pump cavitation, column flooding, and instrumentation drift—often go undetected until they cause failure. Our Fault Detection and Diagnosis services combine first-principles process models with machine learning (classification, PCA, PLS, neural networks) and real-time SCADA/DCS data to detect specific fault signatures, isolate faulty components, classify fault severity, and trigger predictive alerts before operational limits are exceeded, eliminating reliance on manual log reviews or single-variable threshold alarms.

Real applications

Compressor Vibration: Frequency analysis and amplitude trends monitored—detecting bearing wear, impeller imbalance, and surge precursors before mechanical failure.

Heat Exchanger Fouling: Fouling factor and U-value tracked from inlet/outlet T and Q—scheduling cleaning before production cuts.

Pump Cavitation: Suction pressure drop, motor current spikes, and vibration pattern changes identified—avoiding impeller erosion and bearing failure.

Column Flooding: DP trends, tray temperature profiles, and liquid level oscillations monitored—preventing efficiency loss and off-spec product.

Instrument Drift: Cross-correlation of redundant sensors and model-based validation—detecting thermocouple, transmitter, and flow meter drift without manual recalibration.

Soft sensor

Software: ASPEN HYSYS/ UNISIM/ Python

Why?

Your plant never runs with full visibility—critical variables like composition, quality, and efficiency are often unmeasured or too expensive to monitor. Our Soft Sensor services use rigorous process models and real-time data to infer these variables continuously, eliminating the need for laboratory tests, reducing human error, and allowing safe monitoring of toxic or hazardous streams—empowering you to improve quality, reduce costs, and protect your workforce without installing expensive hardware.

Real applications

Air Separation Unit (ASU): Real-time O₂ and N₂ purity inference from column temperatures, pressures, and flows—backing up costly analyzers.

Waste water treatment unit: Online BOD, COD, DO, pH, SVI, & Nutrient Monitoring: Real-time inference of Biological Oxygen Demand, Chemical Oxygen Demand, Dissolved Oxygen, pH, Sludge Volume Index, and Total Nitrogen/Total Phosphorus from process data.

Water unit: Online Silica, Conductivity, TDS, & Resin Breakthrough: Real-time inference from flow, temperature, and bed age—replacing grab samples, conductivity cells, and manual resin testing.

Thank You

High-Fidelity Simulation. High-Confidence Operations.